

Di & Oligo-saccharides

Importance & Function

Polysaccharides

Structure, Classification, & Biomedical Importance



Objectives

At the end of the lecture, students would be able to:

- Know the functions of disaccharides.
- Classify polysaccharides.
- Explain the structure and functions of different polysaccharides.

☐ **Disaccharides: 2 sugar units**

☐ Lactose: Glucose + Galactose

☐ Sucrose: Glucose + Fructose

☐ Maltose: Glucose + Glucose

☐ **Oligosaccharides: 3-9 sugars**

☐ **Polysaccharides: ≥ 10 sugars**

☐ Homopolysaccharide

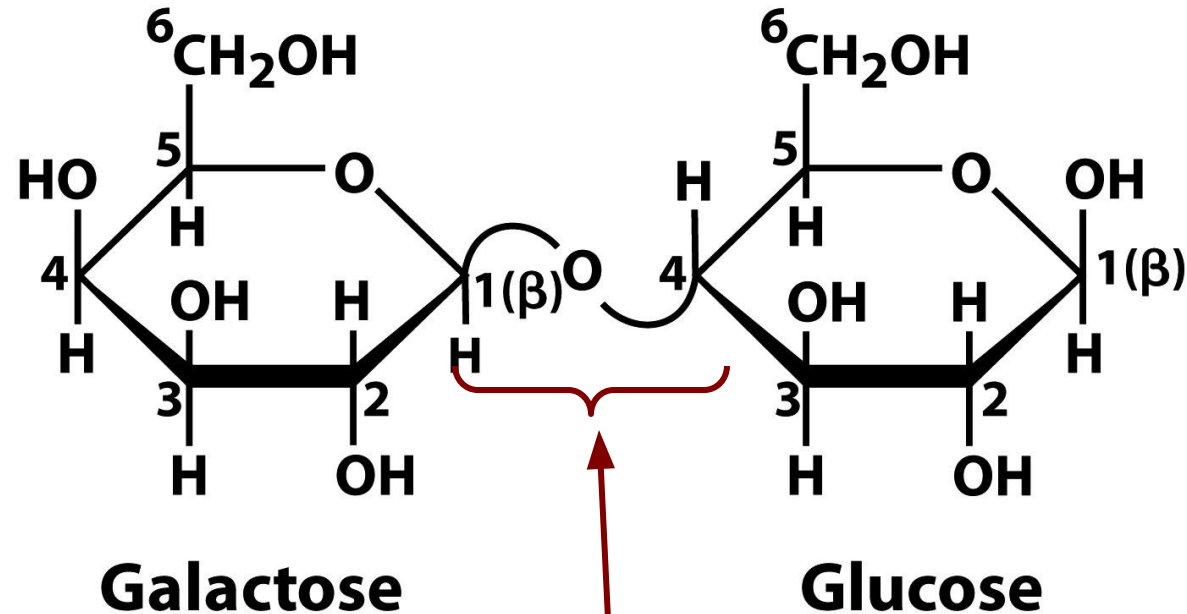
☐ Heteropolysaccharides



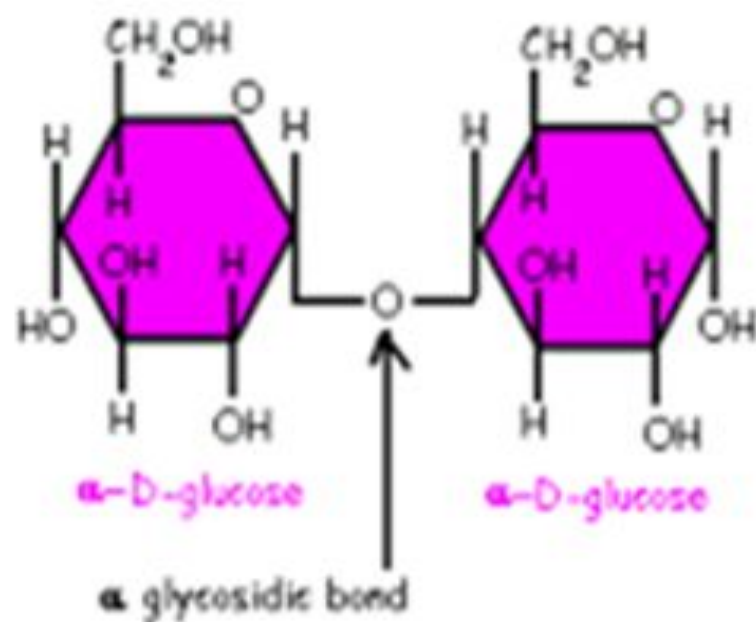
GLYCOSIDIC BOND

Monosaccharides can be linked to form **oligosaccharides**, and ultimately **polysaccharides** via **glycosidic bonds**.

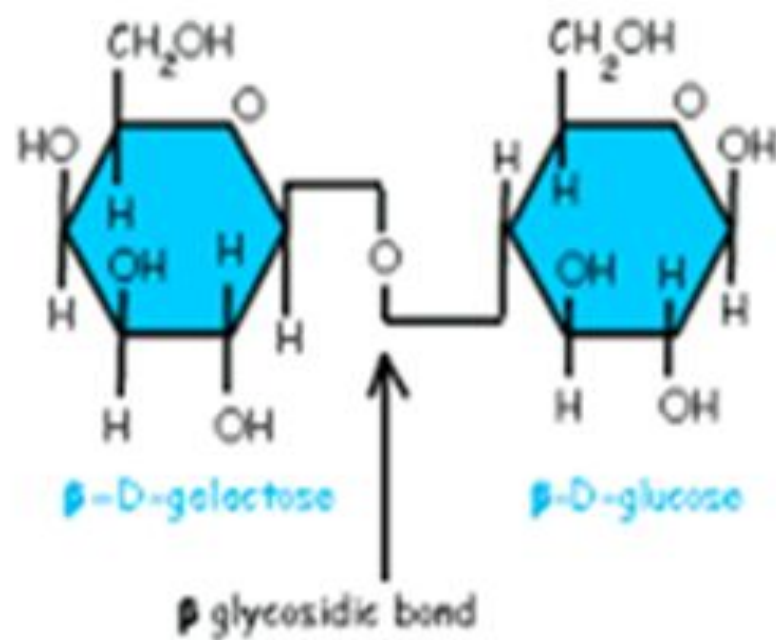
LACTOSE



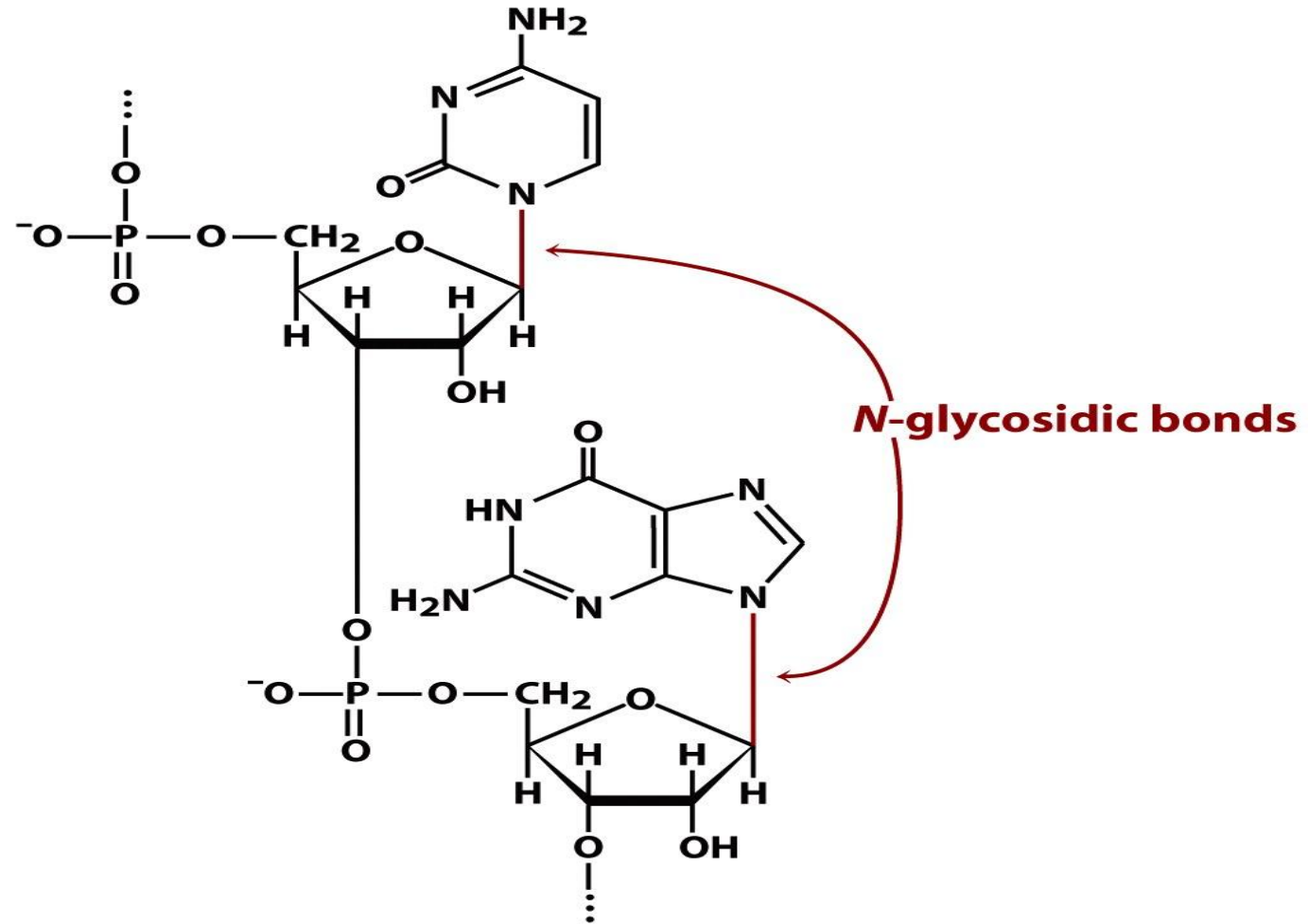
Maltose



Lactose



Glycosidic Bonds in RNA



Disaccharides

- When two monosaccharides are combined together by glycosidic linkage, a disaccharide is formed.

- Two types:-

Non-reducing

Sucrose

Trehalose

Cane sugar

in yeast

Reducing

Lactose

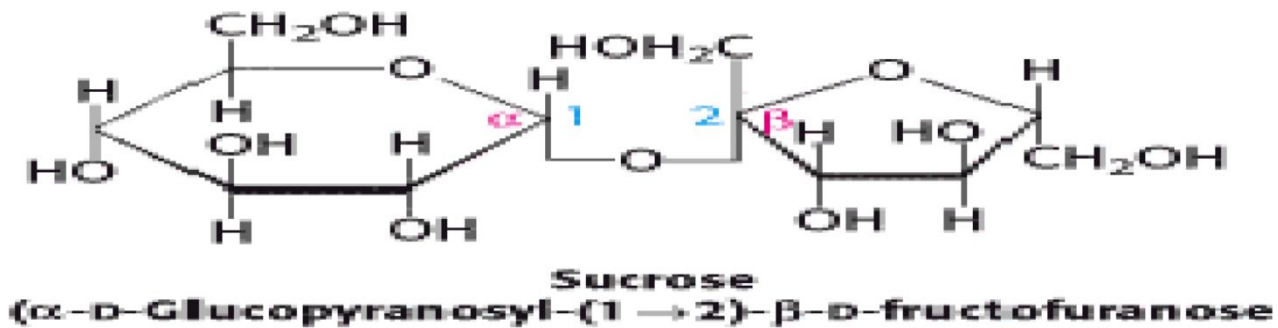
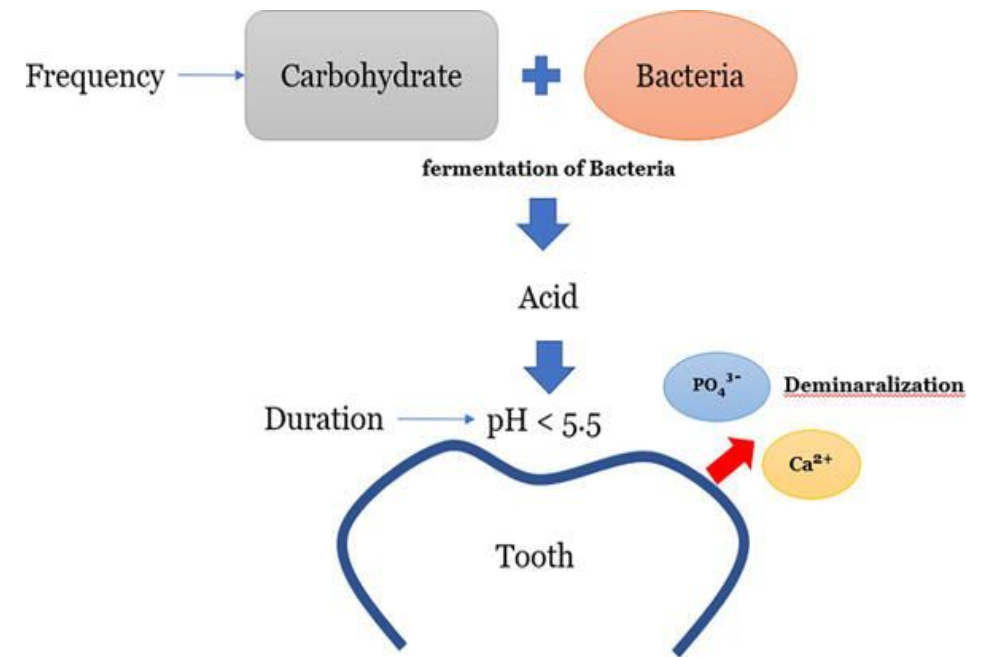
Maltose

Milk sugar

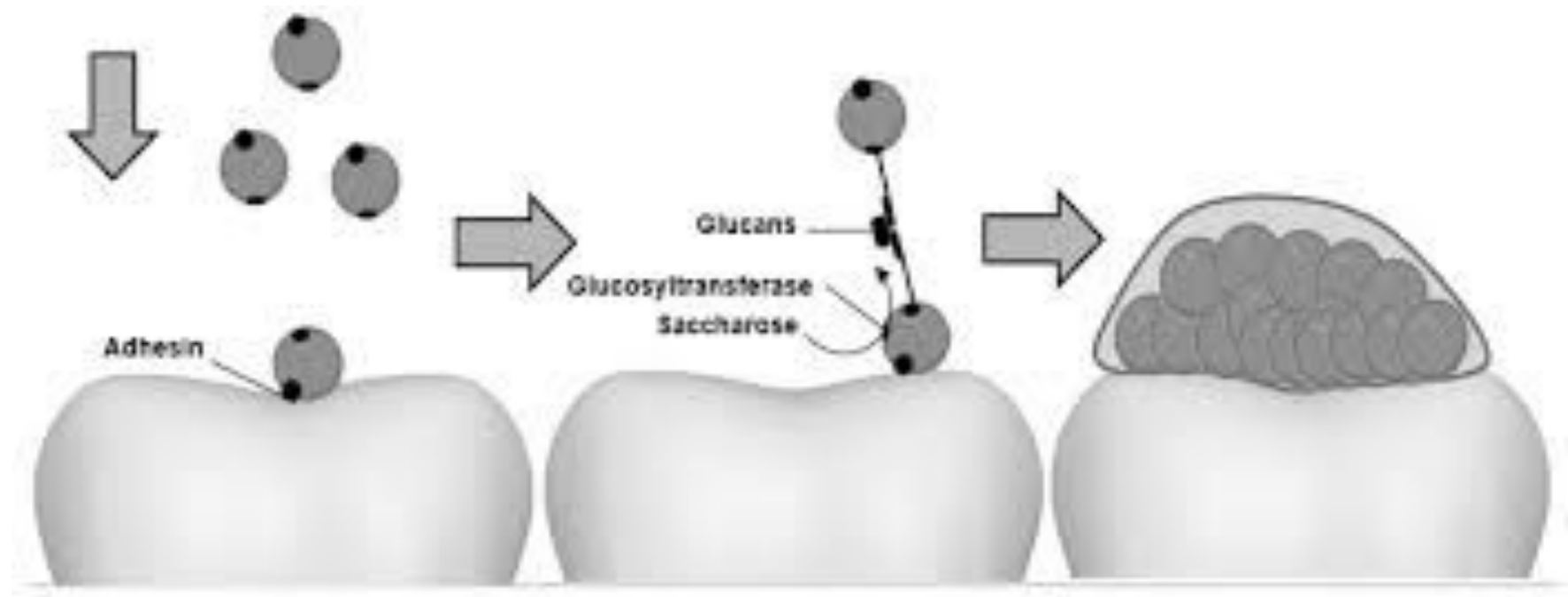
Malt sugar

Sucrose

- Cane sugar, table sugar
- Glu + Fru (1↔2)
- Sweetening agent
- Non-reducing
- No osazones
- **Clinical Importance:-**
 - Dental caries
 - Bypasses metabolic check points- OBESITY
 - “Sucrase deficiency “



- Sucrose is a critical substrate for *Streptococcus mutans* biofilm formation, driving the synthesis of sticky extracellular polysaccharides (EPS).
- It promotes bacterial adhesion to tooth surfaces, creates a protective, dense matrix, and enhances the accumulation of plaque, significantly increasing biofilm virulence.

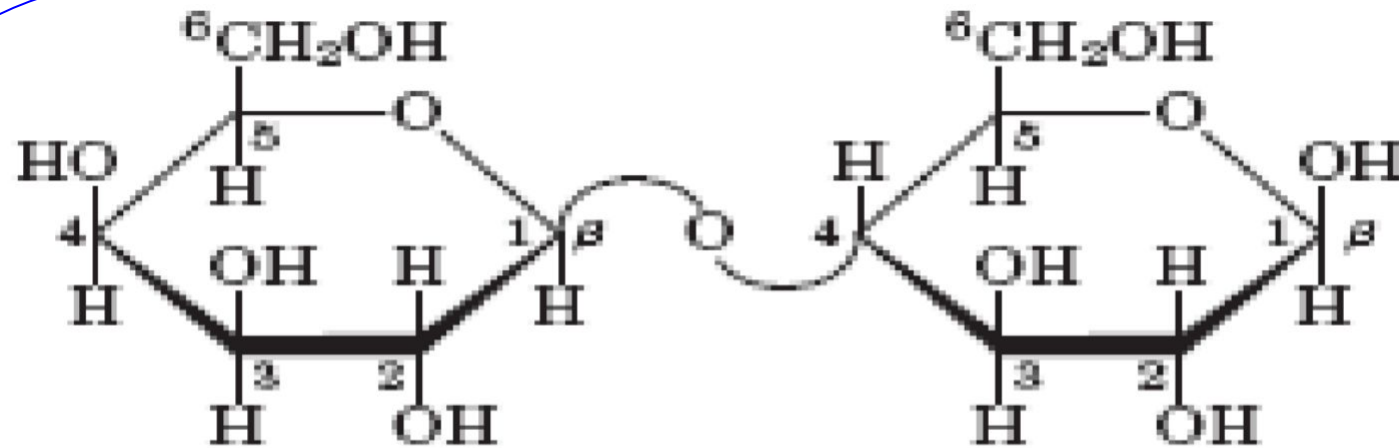


Inversion

- Hydrolysis of sucrose (optical rotation $+66.5^\circ$) will produce one molecule of glucose ($+52.5^\circ$) and one molecule of fructose (-92°)
- Therefore the products will change the dextrorotation to levorotation (INVERSION)
- Equimolecular mixture of glucose and fructose thus formed is called as Invert Sugar
- The enzyme producing hydrolysis of sucrose is called INVERTASE

Lactose

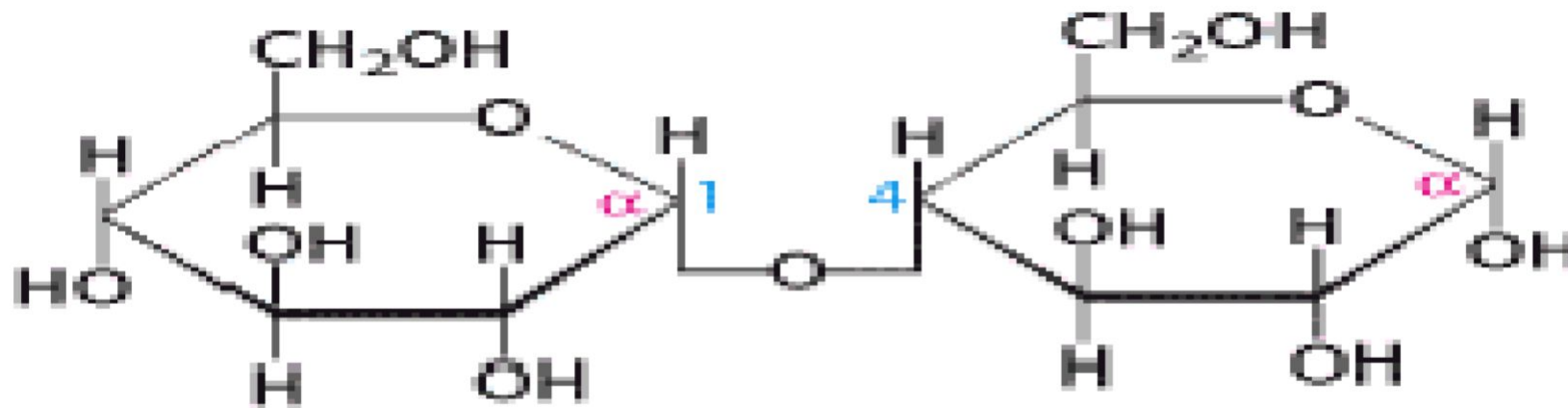
- Milk sugar
- Galactose + Glucose ($\beta 1 \rightarrow 4$)
- Reducing disaccharide
- Beta glycosidic linkage
- Osazone – Powder Puff or hedgehog shaped



Lactose (β form)
 β -D-galactopyranosyl-(1 $\rightarrow$ 4)- β -D-glucopyranose
Gal($\beta 1 \rightarrow 4$)Glc

Maltose

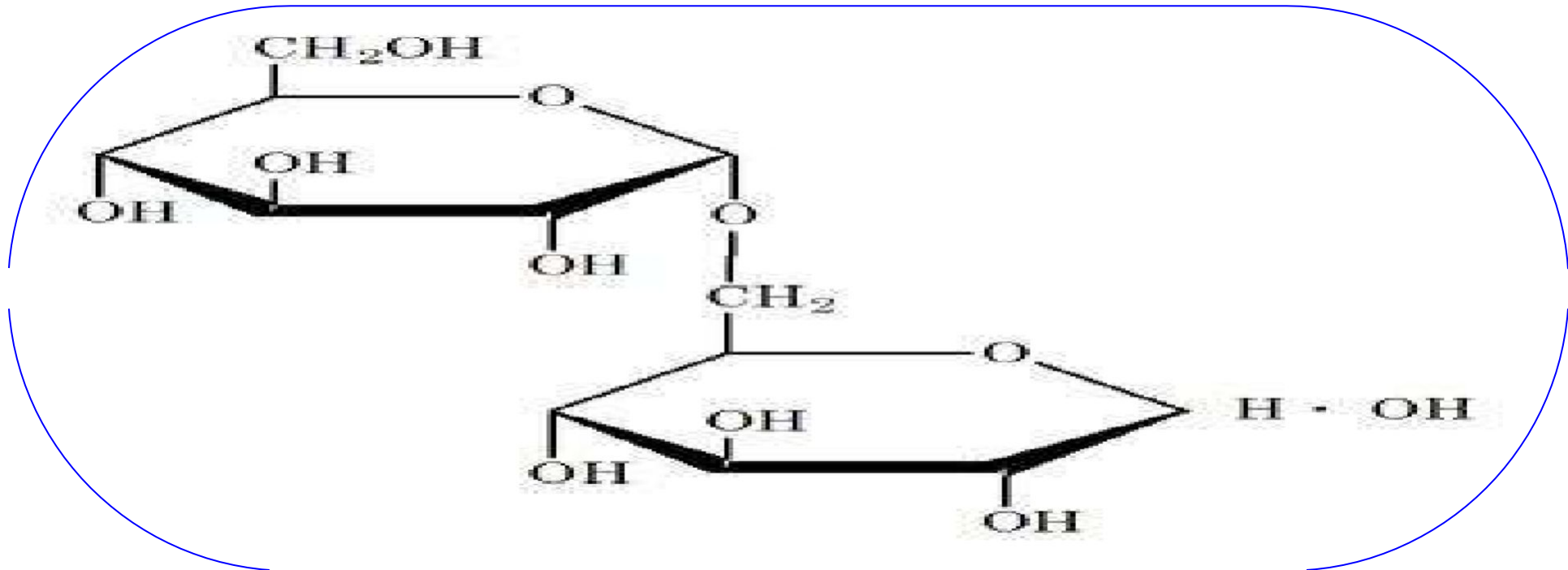
- 2 glucose residues ($\alpha 1 \rightarrow 4$ linkage)
- Reducing disaccharide
- Malt sugar
- Osazone:- Star shaped or flower petal shaped



Maltose
(α -D-Glucopyranosyl-(1 $\rightarrow$ 4)- α -D-glucopyranose)

Isomaltose

- 2 Glucose in α (1 $\rightarrow$ 6) linkage
- Reducing disaccharide
- Produced during partial hydrolysis of starch and glycogen

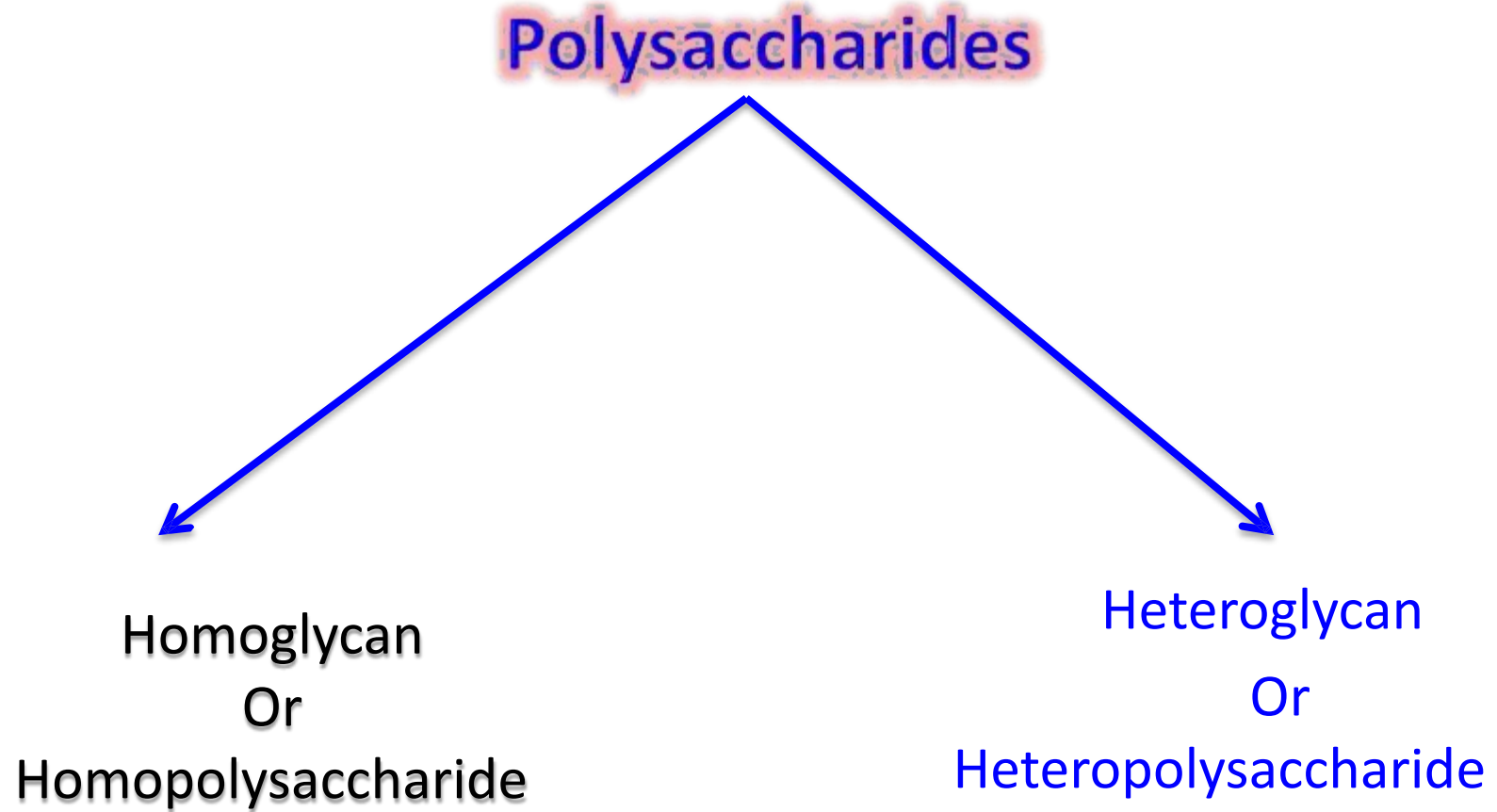


Disaccharides of importance

Sugar	Composition	Source	Clinical Significance
Sucrose	D-glucopyranosyl-(1-2) D-fructo furanoside	Cane and beet sugar, sorghum and some fruits and vegetables	Rare genetic lack of sucrase leads to sucrose intolerance—diarrhea and flatulence
Lactose	D-galactopyranosyl-(1-4)D-glucopyranose	Milk	Lack of lactase leads to lactose intolerance diarrhea and flatulence; may be excreted in the urine in pregnancy

Sugar	Composition	Source	Clinical Significance
Maltose	D-glucopyranosyl-(1-4)-D-glucopyranose	Enzymatic hydrolysis of starch (amylase); germinating cereals and malt	
Isomaltose	D-glucopyranosyl-(1-6)-D-glucopyranose	Enzymatic hydrolysis of starch (the branch points in amylopectin)	
Lactulose	D-galactopyranosyl-(1-4)-D-fructofuranose	Heated milk (small amounts), mainly synthetic	Not hydrolyzed by intestinal enzymes, but fermented by intestinal bacteria; used as a mild osmotic laxative
Trehalose	D-glucopyranosyl-(1-1)-D-glucopyranoside	Yeasts and fungi	

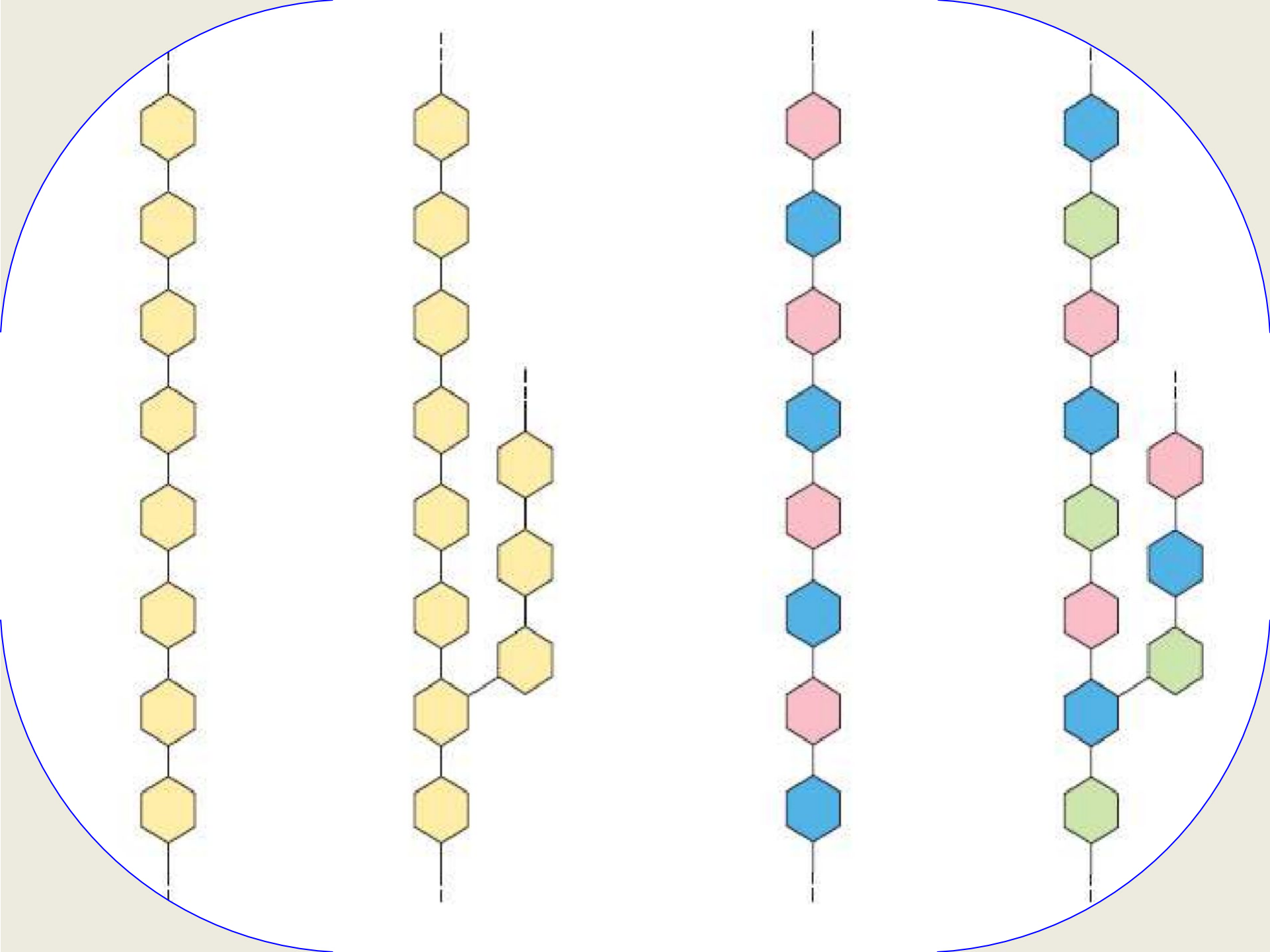
Polysaccharides



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graph TD; A[Polysaccharides] --> B[Homoglycan<br/>Or<br/>Homopolysaccharide]; A --> C[Heteroglycan<br/>Or<br/>Heteropolysaccharide];
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Homoglycan
Or
Homopolysaccharide

Heteroglycan
Or
Heteropolysaccharide



Homoglycans

- Starch
- Glycogen
- Cellulose
- Inulin
- Dextrans
- Chitin

Starch

- Carbohydrates of the plant kingdom
- **Sources:**
- Potatoes, tapioca, cereals (rice, wheat) and other food grains
- Composed of Amylose & Amylopectin
- **Amylose:**
- When starch is treated with boiling water, 10 -20 % is solubilized.
- This part is called amylose, contains glucose units with α -1,4 glycosidic linkages.
- Mol wt =400,000 or more



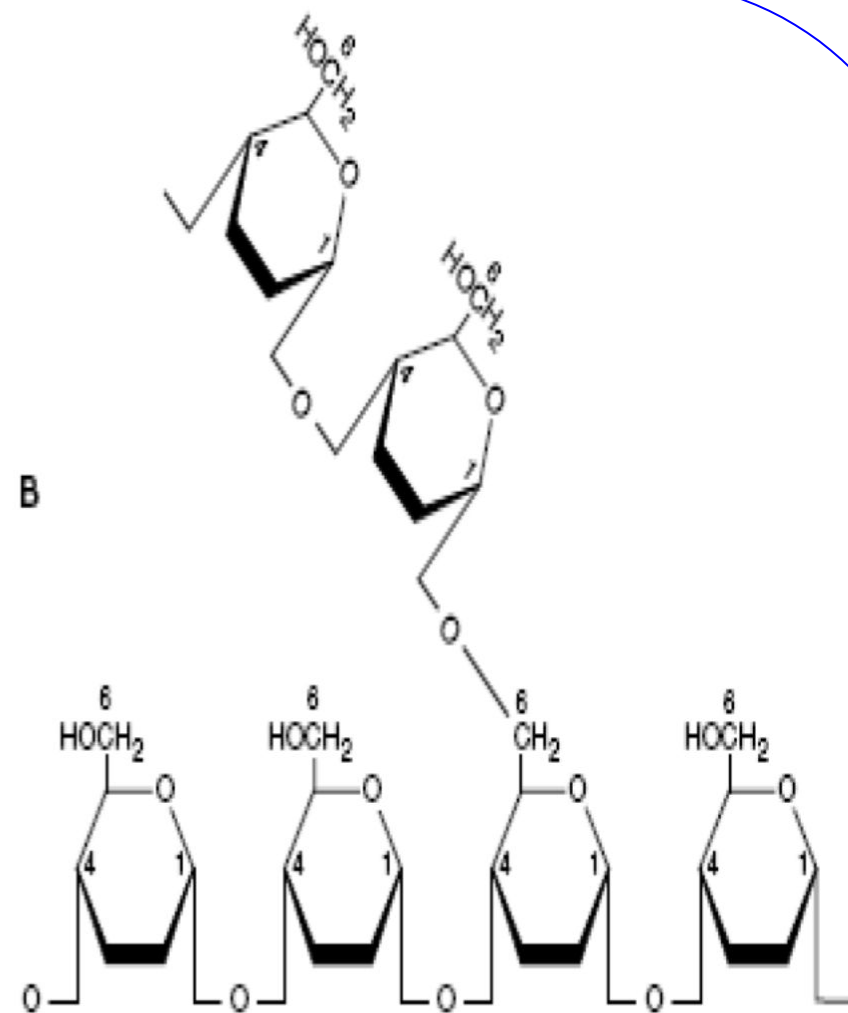
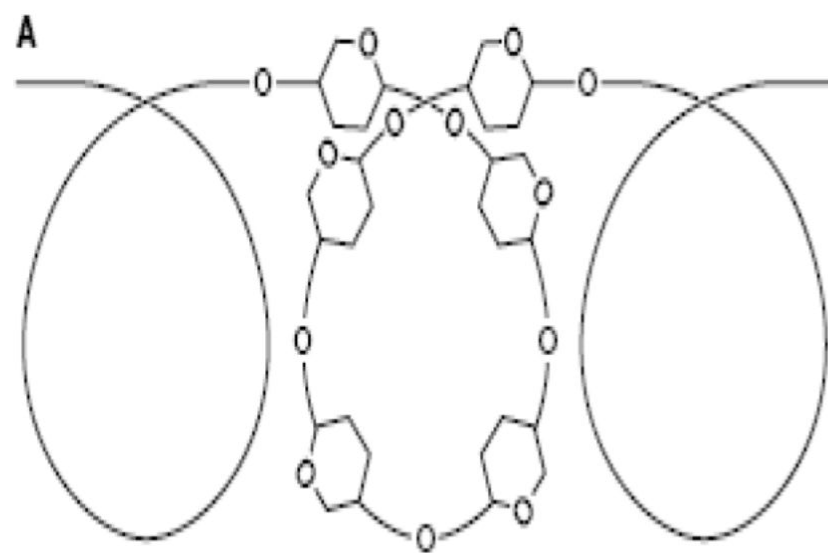
Amylose



Amylopectin

- **Amylopectin:**

- The insoluble part absorbs water and forms paste like gel;
- This is called as amylopectin.
- Amylopectin is also made up of glucose units, but is highly branched with molecular weight more than 1 million.
- The branching points are made by α -1, 6 linkage

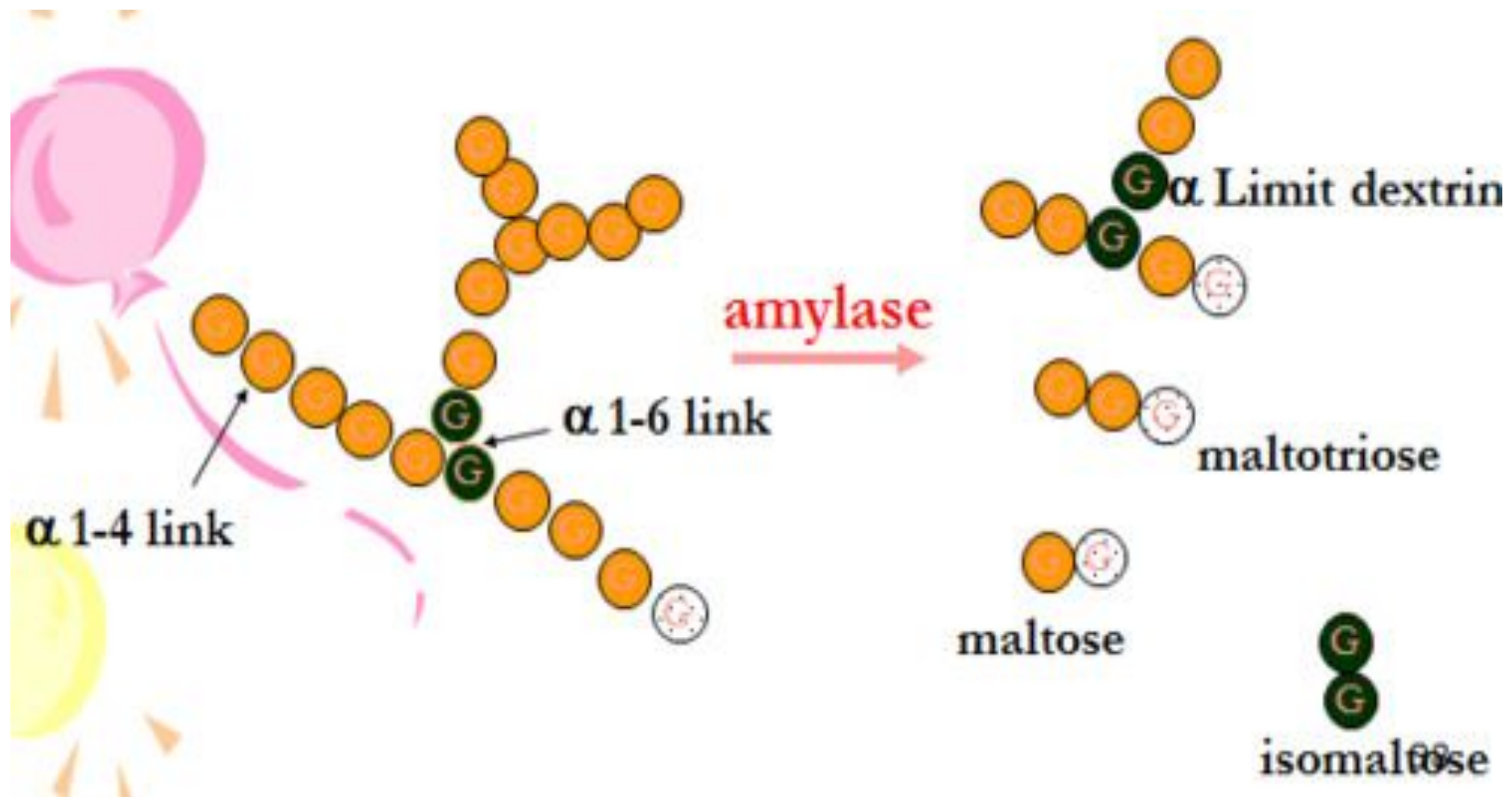


Hydrolysis of starch

- Starch will form a blue coloured complex with iodine.
- This is a sensitive test for starch.
- When starch is hydrolyzed by mild acid, smaller and smaller fragments are produced.

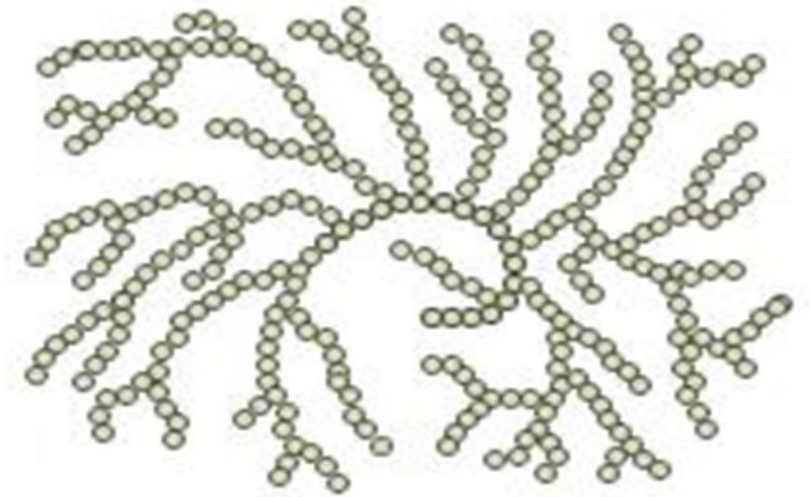
Action of amylases on starch

- Salivary amylases and pancreatic amylases are α amylases, which act randomly on α , 1-4 linkages to split starch into smaller units called dextrins
- Beta amylases (plant origin – almonds etc) act consecutively from one end.
- When beta amylases reach a branch point in amylopectin, enzyme is blocked, leaving a large molecule called as LIMIT DEXTRIN



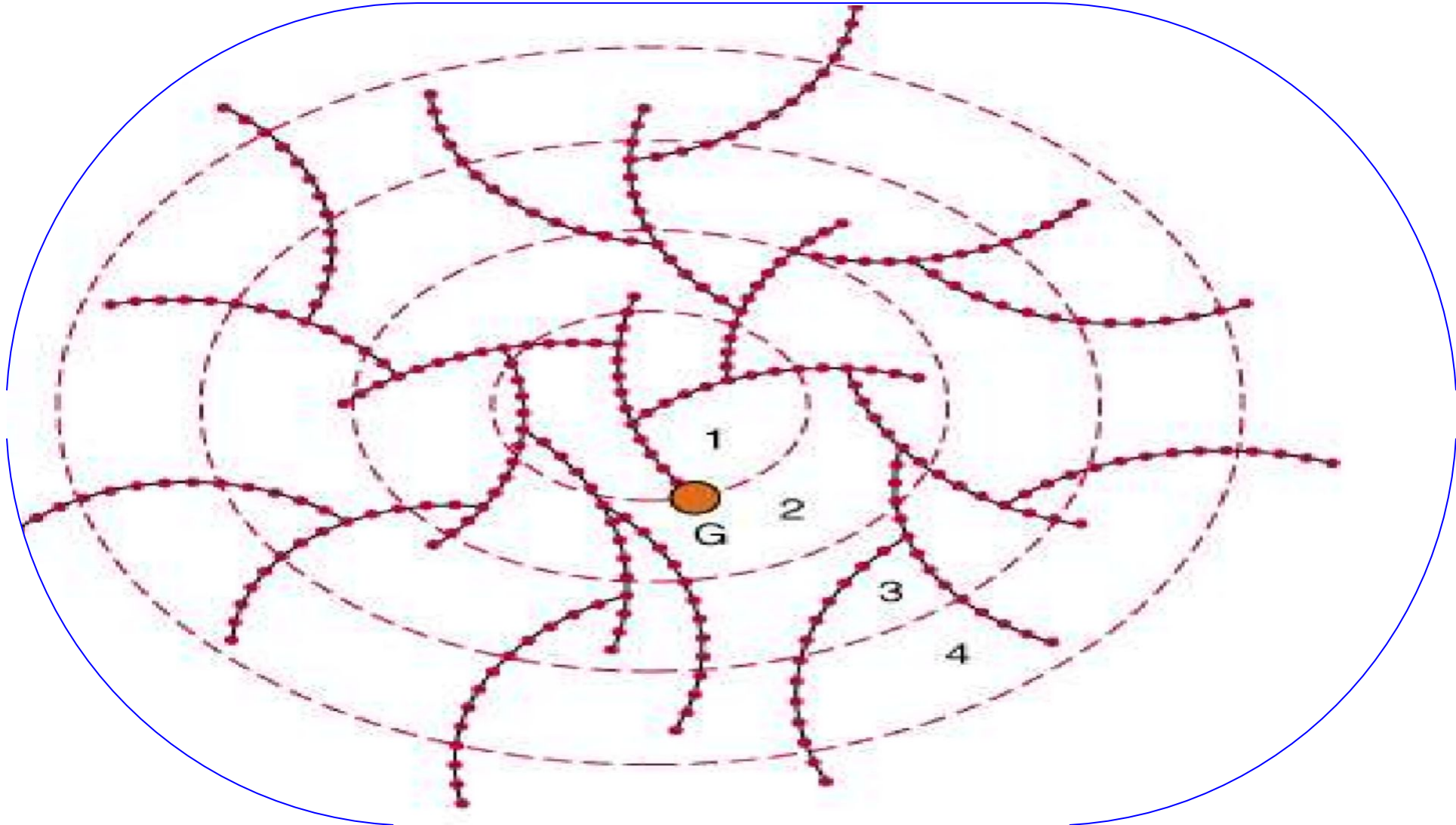
Glycogen

- Storage form of energy in animal.
- Stored in liver and muscle
- Stores more glucose residues per gram than starch.
- More branched and compact than starch.
- Less osmotic pressure.
- More energy in a smaller space.
- Glycogen in liver (about 5–6% of the liver's weight).
- Muscle glycogen represents about 1–2% of total muscle mass.
- Liver glycogen -first line of defense against declining blood glucose levels especially between meals.



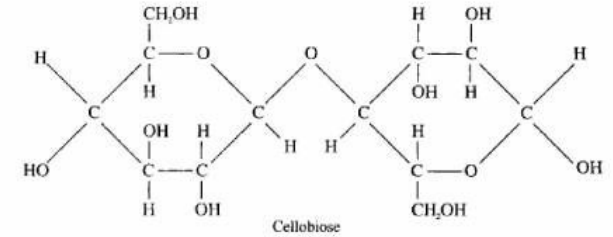
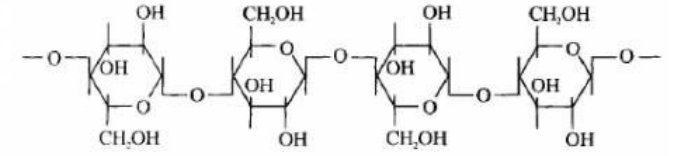
Glycogen

- **A homopolysaccharide:** linear chain of $\alpha(1\rightarrow4)$ linked glucosyl residues with branches joined by $\alpha(1\rightarrow6)$ linkages



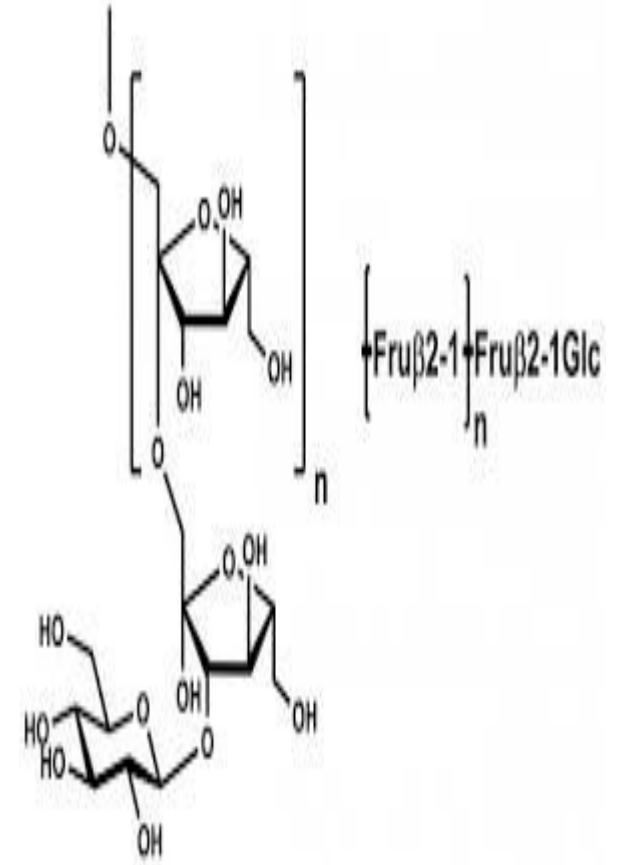
Cellulose

- Glucose units combined by β -1,4 linkages.
- Straight line with no branches.
- Mol wt 2-5 million.
- This bond is digested by **cellobiases** an enzyme not present in humans.
- *Herbivores animals have large caecum which harbor bacteria which break cellulose.*
- White ants (termites) and some wood fungi also have cellulase.
- **Commercial applications:** nitrocellulose, cellulose acetate membranes for electrophoresis ETC



Inulin

- D-fructose in β -1,2 linkages.
- Source:
- Bulbs and tubers chicory, dahlia, dandelion, onions, garlic.
- Not metabolized .
- Not absorbed nor secreted by kidneys.
- USE – to measure GFR.

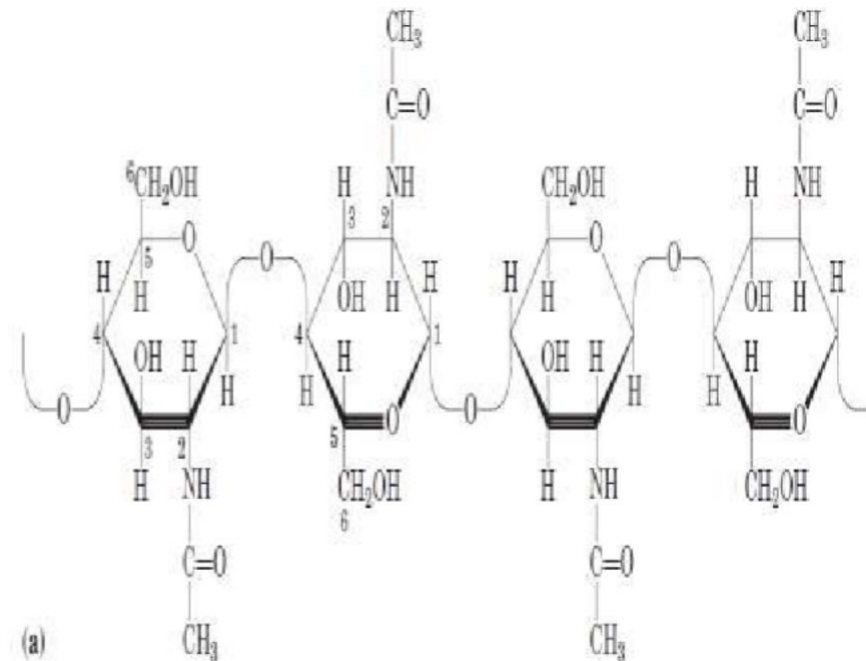


Dextrans

- Highly branched homoglycan containing Glu residues in 1-6, 1-4 and 1-3 linkages.
- Produced by microbes.
- Mol. wt:- 1-4 million.
- As large sized, they will not move out of vascular compartment so used as plasma expanders.

Chitin

- N-acetyl glucosamine with beta 1,4 glycosidic linkage
- Exoskeleton of crustacea and insects.



Thank you